

- 1 The central processing unit of a lap-top computer operates at 3.2×10^9 cycles per second.
 - a Express this frequency in
 - i megahertz
 - ii gigahertz.
 - b Calculate the time period of each cycle.
- 2 When a guitar string was plucked (once) it oscillated with a frequency of 196 Hz.
 - a Determine the *angular* frequency of this oscillation.
 - b Suggest how you would expect the frequency and amplitude to change in the next few seconds. Explain your answer
- 3 a Show that the angular frequency of the Earth spinning on its axis is approximately $7 \times 10^{-5} \text{ rad s}^{-1}$.
 - b Determine the total angle (rad) through which it will rotate in 100 hours.
- 4 A car engine was measured to have 3755 rpm (revolutions per minute). Calculate its angular frequency.
- 5 Using slow-motion video replay, the angular frequency of the oscillations of a humming bird's wings was found to be 272 rad s^{-1} . Determine how many times it beat its wings in one minute.

Two commonly investigated oscillators

SYLLABUS CONTENT

- The time period of a mass–spring system as given by:

$$T = 2\pi \sqrt{\frac{m}{k}}$$

- The time period of a simple pendulum as given by:

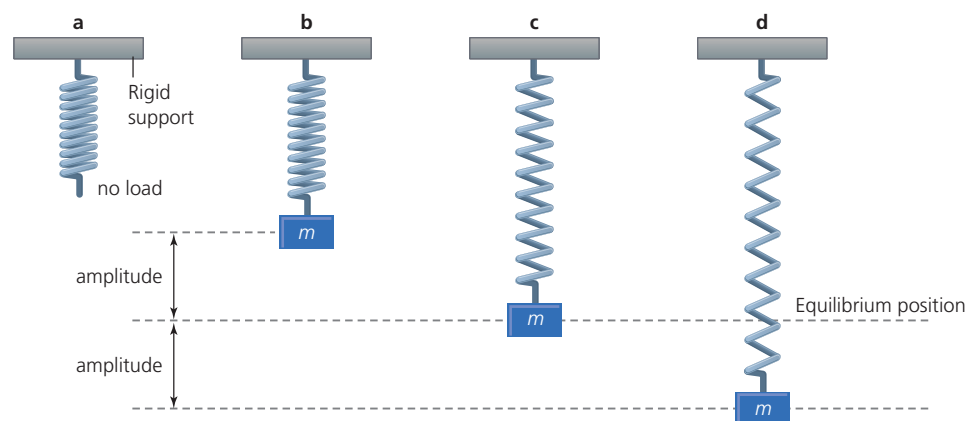
$$T = 2\pi \sqrt{\frac{l}{g}}$$

A mass–spring system and a simple pendulum are important for a number of good reasons. These include:

- The proportional relationships between force and displacement are easily understood.
- Their periods of oscillation have a convenient time for measurement in a school laboratory.
- They are good approximations to simple harmonic motion.
- Energy is dissipated slowly, so that the oscillations continue for a long enough time that allows for accurate measurements to be completed.
- They act as starting points for understanding many similar, but more complex, oscillators.

■ Mass-spring system

We have already briefly discussed a mass oscillating horizontally between two springs (Figure C1.2). A more common arrangement is shown in Figure C1.5. Of course, in this arrangement, the force of gravity (the weight of the mass) also acts vertically, but it does not affect the results because it is constant. The system would have the same time period if it was moved to a location where the gravitational field strength was different.

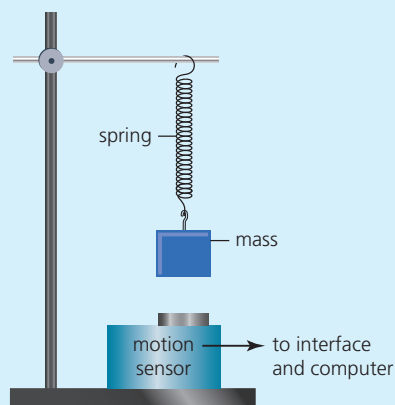


■ **Figure C1.5** Different positions of a mass, m , oscillating on a spring

Tool 2: Technology

Use sensors

Position sensors are useful in many aspects of the study of motion, including mechanical oscillations. Figure C1.6 shows how digital data can be collected, which will be processed later by the computer.



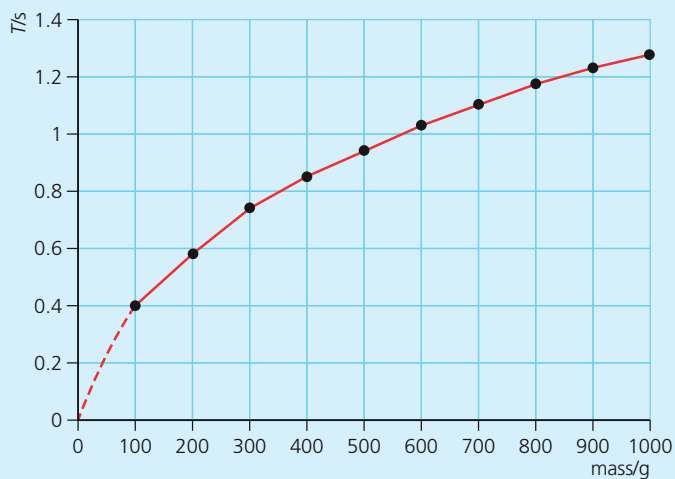
■ **Figure C1.6** Using a sensor to investigate oscillations of a mass on a spring.

We will assume that the spring is never overstretched and that it obeys Hooke's law (see Topic A.2):

$$F_H = -kx$$

Or, in words: the size of the restoring force acting on the mass = spring constant $\times$ displacement from its equilibrium position. The spring constant is a measure of the spring's stiffness (= force / deformation).

Laboratory investigations of the time periods of a mass on a spring are straightforward, especially those involving using different masses on the same spring. For typical results see Figure C1.7. Different springs, or combinations of springs, used with the same mass, can be used to investigate the effect of the spring constant, k , on the period.



■ **Figure C1.7** Variation of time period, T , with mass, m , on a spring

Tool 3: Mathematics

Linearize graphs

You may have learnt how to linearize graphs in Topic B.1. The graph seen in Figure C1.6 appears that it might represent the relationship $T^2 \propto m$. Check this by using information from the graph to draw a T^2 - m graph (uncertainties in T are ± 0.05 s). It should produce a straight line through the origin. Use the graph and the relationship shown below to determine a value for the spring constant, k .

The exact relationship for the SHM of a mass on a spring is as follows:

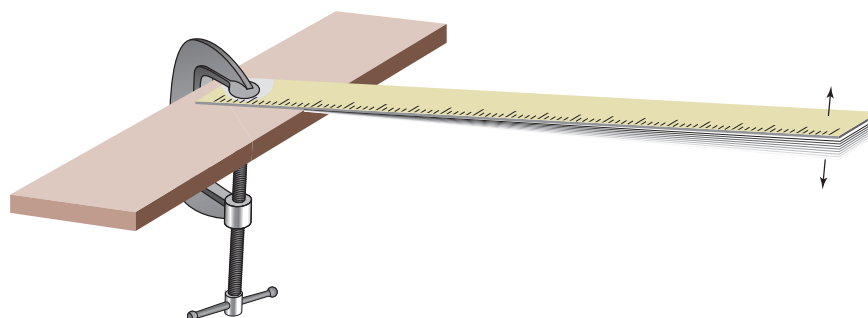
Time period of a mass-spring system:



$$T = 2\pi \sqrt{\frac{m}{k}}$$

Since $\omega = \frac{2\pi}{T}$, this equation can also be written as: $\omega^2 = \frac{k}{m}$.

This equation applies perfectly only to a point mass acted upon by a separate simple spring system that has a well-defined stiffness (k) and no (significant) mass. There is an extremely large number of other oscillating mechanical systems that have similarities to this simple model but are much more complex. For example, oscillations in buildings and bridges. Figure C1.8 shows a simpler example, in which the mass is spread uniformly along the oscillating system, a ruler.



■ Figure C1.8 Oscillating ruler

Inquiry 3: Concluding and evaluating

Concluding

A student wants to investigate an oscillating ruler, similar to that seen in Figure C1.8, but she varied the mass of the ruler by taping various masses near to its free end. She predicted that the time period of the oscillator could be determined from the same equation as for a mass oscillating vertically on the end of a spring. Table C1.1 summarizes her raw data. Assume that the uncertainties are low.

■ Table C1.1 Results of vibrating ruler experiment

Mass on end of blade / g	Time period / s
0	too quick to measure
40	0.55
80	0.76
120	0.91
160	1.04
200	1.15

Process the results and reach a conclusion. State whether the student predicted correctly. If not, suggest a possible reason.

WORKED EXAMPLE C1.2

A 200 g mass was placed on the end of a long spring and increased its length by 5.4 cm.

- Determine the spring constant, k , of the spring.
- If the mass is displaced a small distance from its equilibrium position and undergoes SHM, calculate the frequency of oscillations.
- Suggest a possible reason why oscillations with greater amplitude may not be simple harmonic.

Answer

a $F_H = -kx$

$$0.200 \times 9.8 = -k \times (5.4 \times 10^{-2})$$

$k = -36 \text{ N m}^{-1}$ (the negative sign shows that force and displacement are in opposite directions)

b $T = 2\pi\sqrt{\frac{m}{k}} = 2 \times \pi \times \sqrt{\frac{0.200}{36}} = 0.47 \text{ s}$

$$f = \frac{1}{T} = \frac{1}{0.47} = 2.1 \text{ Hz}$$

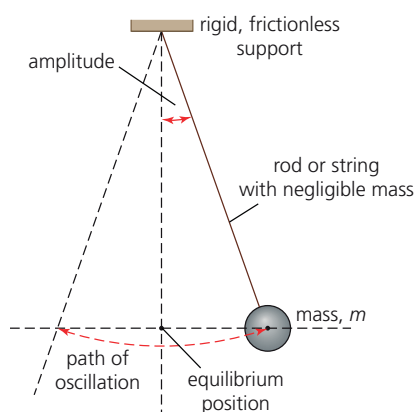
- c** The spring may become overstretched, so that Hooke's law is no longer applicable.

Simple pendulum

◆ **Pendulum** A weight, which is suspended below a pivot, which is able to swing from side to side. The weight is sometimes called the **pendulum bob**. The concept of a **simple pendulum** is a point mass on the end of an inextensible string.

There are many different designs of **pendulum**, all of which involve a mass swinging from side to side under the effects of gravity. We use the term '**simple pendulum**' to describe the simplest possible model of a pendulum: a spherical mass swinging on a rod, or string (both of which have negligible mass), from a rigid, frictionless support. See Figure C1.9. The mass is often called a **pendulum 'bob'**.

Because the mass is spherical and the rod / string has negligible mass, all of the mass of the pendulum can be assumed to be acting as a point mass at the centre of the sphere. The displacement of the motion can be considered to be the angle between the rod / string and the vertical. The restoring force is provided by gravity: consider Figure C1.10.

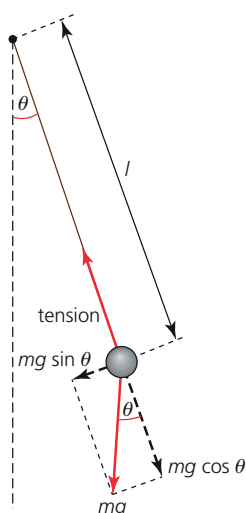


■ **Figure C1.9** Simple pendulum

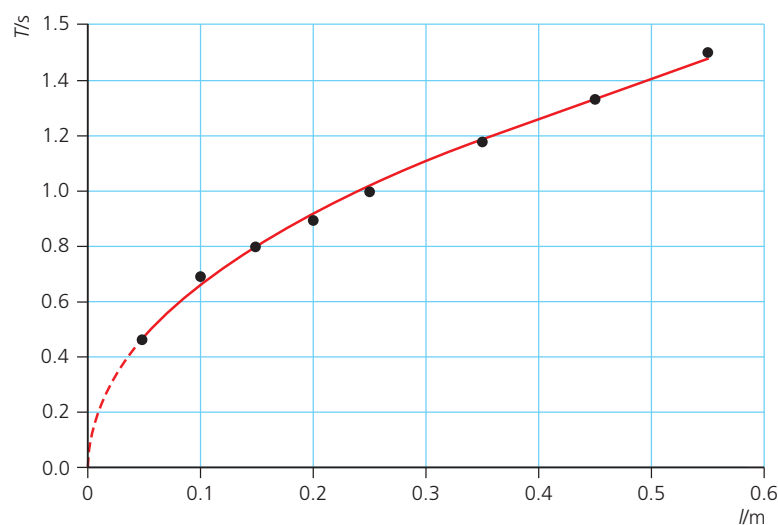
The weight, mg , of the pendulum is conveniently resolved into two perpendicular components: a force $mg \sin \theta$ provides the restoring force bringing the pendulum back towards its equilibrium position; a force $mg \cos \theta$ keeps the rod / string in tension.

An alternative way of looking at this situation: the pendulum has only two forces acting on it: tension in the rod / string and weight, mg . The resultant of these is the restoring force, $mg \sin \theta$. Practical laboratory investigations of a simple pendulum will confirm that, *for small amplitudes*, the time period, T , depends only its length, l (measured from its point of support to its centre of mass). See Figure C1.11.

Period is *not* affected by its mass. This is because doubling the mass, for example, will also result in doubling the restoring force.



■ **Figure C1.10** Components of the weight of the pendulum



■ **Figure C1.11** Results of simple pendulum investigation

The exact relationship for the SHM of a simple pendulum (for small amplitudes) is as follows.

Time period of a simple pendulum:



$$T = 2\pi \sqrt{\frac{l}{g}}$$

Top tip!

The data of Figure C1.11 can be linearized. A graph of T^2-l should be a straight line through the origin. A simple pendulum experiment can be used to determine a value for the gravitational field strength, g . The graph of T^2-l will have a gradient of: $4\pi^2/g$

Tool 3: Mathematics

Use approximation and estimation

For the motion of a simple pendulum to be a good example of SHM, we require that the restoring force ($mg \sin \theta$) is proportional to the displacement (θ). For very small angles, values of $\sin \theta$, $\tan \theta$ and θ (in rad) are almost identical, so that this condition is satisfied. But, as the angle increases, the difference gets greater. Determine the minimum angle for which there is at least a 1% difference between $\sin \theta$, $\tan \theta$ and θ (in rad).



■ **Figure C1.12** Mount Nevado Huascarán in Peru is reported to have the lowest gravitational field strength on Earth; its summit (6768 m) is one of the farthest points on the surface of the Earth from Earth's centre

ATL C1A: Research skills, thinking skills

Use search engines effectively; providing a reasoned argument to support conclusions

The equation highlighted above shows that the period of a simple pendulum would also change if its value was checked in a location where the gravitational field strength had a different value. Using your own research, investigate the variations of the strength of the gravitational field around the Earth. Then state and explain whether experiments in school laboratories would be able to measure those differences.